From Orbit to Device: Spectrum Intelligence for Multi-Tier LEO-Aided D2C IoT Networks

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***Abstract*—The rapid expansion of the Internet of Things (IoT) demands pervasive, all-terrain connectivity that traditional terrestrial networks cannot fulfill alone. To improve the spectral efficiency of both User Equipment (UE) and IoT devices, we propose a multi-tier Direct-to-Cell (D2C) satellite network consisting of multiple constellations at different altitudes in space. To contextualize this paradigm shift, we first outline the architectural evolution from traditional Non-Terrestrial Networks (NTN) to emerging multi-tier D2C ecosystems. Building on this foundation, we introduce a ground-controller-assisted Spectrum Intelligence layer. Rather than relying on rigid association policies, this centralized orchestrator dynamically coordinates connectivity across varying orbital altitudes. The framework executes a dynamic, two-stage routing protocol: first, it performs an intelligent constellation selection to route devices between a lower-altitude capacity tier and a higher-altitude coverage tier; second, it executes a greedy satellite selection within the chosen tier to optimize link quality. Through extensive system-level simulations, we demonstrate that this network-side intelligence successfully mitigates inter-satellite interference and improves spectral efficiency for pervasive IoT applications. The proposed framework achieves about 53% increase in resource-normalized spectral efficiency for broadband UEs and 63.6% increase for IoT devices compared with the naive static multi-tier baseline (BL3), which assigns all UEs to the 550 km capacity tier and all IoT devices to the 1200 km coverage tier.**

***Index Terms*—Spectrum Intelligence, Direct-to-Cell (D2C), Low Earth Orbit (LEO), Internet of Things (IoT), Multi-Tier.**

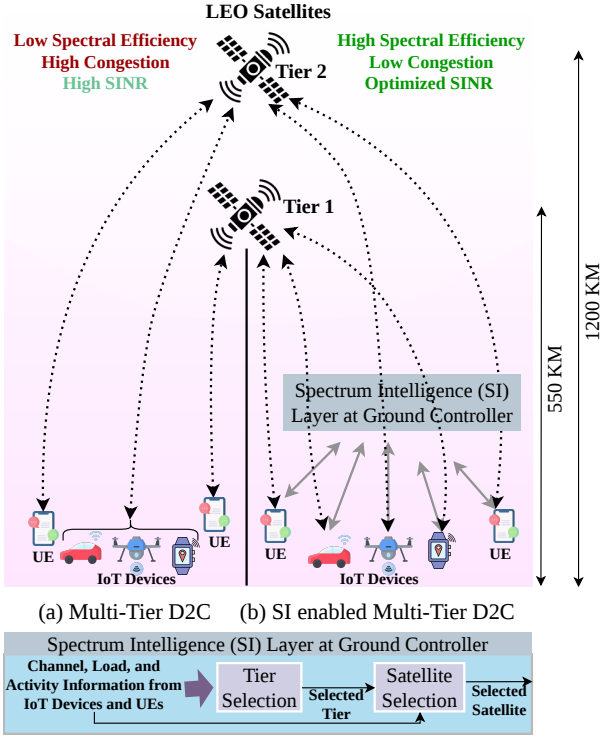


Fig. 1: Conceptual overview of the proposed architecture. (a) A baseline multi-tier D2C network suffering from load imbalances, spectrum congestion, and sub-optimal link efficiency. (b) The proposed multi-tier D2C framework utilizing a spectrum intelligence layer at the ground controller, which dynamically utilizes device telemetry to optimize Tier and Satellite selection.

I. Introduction

The realization of a pervasive, globally interconnected Internet of Things (IoT) is a foundational objective of the impending 6G era. As the demand for ubiquitous, unbroken connectivity scales beyond the geographic constraints of terrestrial infrastructure, the telecommunications industry is increasingly leveraging space-based networks. Recognizing this paradigm shift, the 3rd Generation Partnership Project (3GPP) has heavily focused its recent standardization efforts (specifically within Releases 17 and 18) on integrating Non-Terrestrial Networks (NTN) into the 5G ecosystem [1], [2]. However, traditional NTN architectures rely on complex relay chains requiring specialized, high-gain ground gateways to distribute signals. This inherently limits the deployment of low-power, lightweight IoT devices. In response, the industry has pivoted toward emerging Direct-to-Cell (D2C) architectures, which unlike traditional NTN, eliminate the intermediary ground gateways, allowing satellites to communicate *directly* with consumer smartphones and off-the-shelf IoT devices [3], [4].

Despite the promise of "out-of-the-box" global connectivity, current D2C architectures are largely built around single-tier Low Earth Orbit (LEO) constellations, which face major limitations when serving power-constrained IoT devices over congested sub-6 GHz spectrum. These systems must cope with Doppler variation, localized RF congestion, interference, and the conflicting requirements of low latency and wide-area coverage [5]. To address these limitations, this article proposes a *multi-tier D2C satellite network* architecture in which multiple LEO tiers operate at distinct altitudes. By distributing devices and spectrum demand across complementary orbital tiers, the proposed framework aims to improve spectral efficiency, network capacity, and coverage reliability for both standard User Equipment (UE) and massive IoT deployments [6].

The Bottleneck: Spectrum Sharing in LEO-D2C Networks. Despite the potential promise of multi-tier D2C architectures, integrating thousands of highly mobile LEO satellites with existing terrestrial frameworks exposes a critical vulnerability: spectrum scarcity. Because D2C networks must communicate

directly with ground devices, they are forced to operate within existing, highly congested sub-6 GHz mobile spectrum bands (e.g., LTE Band 41 and Band 7). This spectrum reality pushes the physical layer to its absolute limits. A LEO satellite orbiting between 500-2000 km moves at immense speeds, subjecting narrowband signals to severe Doppler shifts, atmospheric attenuation, and rapid Line-of-Sight (LOS) transitions.

As multiple UEs and IoT devices attempt to access the network, they must reuse or compete for the same time-frequency-beam resources, co-channel spectrum, and subbands within or across LEO constellations. This creates a severe spectrum-sharing problem. When a satellite footprint sweeps over a densely populated urban center, the resulting spike in device density, combined with inter-satellite interference from neighboring constellations, precipitously degrades the Signal-to-Interference-plus-Noise Ratio (SINR) for both uplink and downlink transmissions [7]. Without intervention, this drop in Spectral Efficiency (SE) threatens to render D2C links unusable in the exact scenarios where they are most needed.

The Challenge: Constrained IoT Devices. This degradation is especially critical for low-power IoT devices because their limited transmit power, antenna gain, and processing capability make long-distance D2C links more vulnerable to poor SINR. Narrowband Internet of Things (NB-IoT) and Long-Term Evolution for Machines (LTE-M) devices may operate at maximum transmit powers of 20–23 dBm and employ omnidirectional antennas with 0 dBi gain, as considered in the 3GPP IoT-NTN reference scenarios [8]. Under the considered 2 GHz carrier frequency and 40° elevation angle, the corresponding free-space path losses are approximately 157 dB and 163 dB for the 550 km and 1200 km shells, respectively. Furthermore, the limited processing capability and energy budget of IoT devices preclude complex constellation-selection algorithms at the device level. This constraint directly motivates the proposed network-side architecture, in which all channel estimation, load monitoring, and shell-assignment decisions are handled entirely at the ground controller, placing no additional computational or signaling burden on the IoT device.

The Paradigm Shift: Spectrum Intelligence. To prevent the spectrum bottleneck from stifling IoT scalability, the telecommunications paradigm must shift from traditional static spectrum allocation to active *Spectrum Intelligence*. We propose to realize this shift by integrating a ground-controller-assisted intelligence layer into the multi-tier LEO-D2C network. This centralized orchestrator aggregates real-time device telemetry, including channel states, load distributions, and activity patterns [9], and uses this information to coordinate device-to-tier association and in-band spectrum sharing for both broadband UEs and hardware-constrained IoT devices. In this work, *Spectrum Intelligence* denotes a centralized, context-aware, rule-based spectrum optimization mechanism; which may also utilize advanced algorithmic methods such as artificial intelligence or machine learning.

Fig. 1 illustrates this paradigm shift through two subplots. Fig. 1a shows the conventional multi-tier deployment, where devices remain attached to fixed orbital tiers and therefore experience load imbalance, localized congestion, and inefficient spectrum use. In contrast, Fig. 1b shows the proposed Spectrum Intelligence framework, where a ground-controller-assisted intelligence layer performs two-stage connectivity management: first selecting the appropriate orbital tier and then selecting the serving satellite within that tier based on SINR. Rather than forcing devices to remain tethered to fixed LEO tiers, the intelligence engine dynamically orchestrates tier association and in-band coexistence to improve the use of limited D2C spectrum.

Scenario of Interest: We analyze a highly dynamic, multi-tier satellite environment serving a dense deployment of heterogeneous UEs and IoT devices. The communication channel is subjected to both deterministic attenuation (tropospheric/ionospheric loss, path loss) and stochastic effects (shadowing, Rician fading). We focus our experimental evaluation on the dense urban topology of Tokyo. As the LEO constellations traverse this high-density region, the rapid orbital movement generates a highly volatile channel, interference, and load telemetry. The spectrum intelligence layer utilizes this data to dynamically update device-to-tier associations to maximize downlink SE even under severe urban congestion.

Summary of Contributions.

- We provide a detailed landscape of the evolution from traditional NTN to emerging D2C architectures, formally introducing the concept of a *multi-tier LEO-D2C network* to overcome the spectral and coverage limitations of current single-tier systems (Sec. II).
- We propose a *spectrum intelligence layer*, deployed in a Ground Controller unit, which dynamically couples intelligent constellation tier selection with greedy satellite selection to maximize device-level SINR and optimize both uplink and downlink resources against RF congestion (Sec. III).
- We conduct system-level simulations to provide a comparative analysis of network throughput across various geographic locations and user densities, definitively demonstrating that ground-controller-assisted intelligent constellation allocation is a strict prerequisite for mitigating interference and ensuring resilient IoT coverage in the 6G era [6] (Sec. IV).
- We identify critical open challenges and outline future research directions necessary for scaling the spectrum intelligence layer globally, highlighting key technological vectors such as collaborative edge Artificial Intelligence (AI), inter-constellation coexistence protocols, AI-assisted beam tracking, predictive cross-tier handovers, and the integration of 3D orbital digital twins (Sec. V).

II. Evolution to Multi-Tier D2C Networks

The quest for ubiquitous global coverage is not new. The architectural mechanisms proposed to achieve this goal have evolved dramatically over the past decade. To fully appreciate the necessity of a ground-controller-assisted spectrum intelligence layer, one must first understand the fundamental architectural shift from traditional NTN to emerging multi-tier D2C ecosystems. To provide a consolidated overview of these

technological paradigms, Table I systematically contrasts traditional NTN, single-tier conventional D2C, and the proposed multi-tier D2C framework across key hardware requirements, link performance metrics, and physical channel impairments.

The Traditional NTN Paradigm: Historically, satellite communication networks were conceptualized strictly as backhaul extensions of terrestrial cellular infrastructure. Under early 3GPP paradigms, satellites, whether Geostationary (GEO) at 35,786 km, Medium Earth Orbit (MEO), or LEO, functioned primarily as "bent-pipe" relays or remote macro-cells in space [1].

The architecture of a standard NTN involves a complex relay chain: the signal travels from the satellite to a specialized, high-gain ground gateway, such as a Very Small Aperture Terminal (VSAT), before being distributed locally to UEs. Because the ground gateway is equipped with sophisticated tracking antennas (often exceeding 30 dBi in gain) and high-power amplifiers (transmitting at 10 W to 50 W), it can easily compensate for the massive free-space path loss, often exceeding 160 dB in sub-6 GHz bands, and atmospheric attenuation inherent in space-to-ground links [5].

Consequently, traditional NTNs can achieve broadband data rates. However, this architecture severely limits the vision of an "all-pervasive" IoT. Requiring specialized, expensive, and power-hungry relay hardware completely precludes the integration of IoT devices at scale such as Narrowband IoT (NB-IoT) or enhanced Machine Type Communication (eMTC) sensors. These devices are constrained by maximum transmit powers of 23 dBm (200 mW), zero-gain omnidirectional antennas, and strict 10-year battery life targets [8].

The D2C Revolution and Multi-Tier Constellations: The D2C framework represents a radical departure from the relay-centric NTN model. Recent NTN research also considers regenerative satellite payloads as an alternative to transparent *bent-pipe* architectures. Regenerative systems place full or partial gNB functions onboard the satellite, enabling local signal processing, routing, and more flexible radio-resource management [2], [5]. These capabilities can reduce dependence on the feeder link and support adaptive bandwidth, beam, and power allocation. However, they also impose greater onboard processing, power, thermal, and payload-complexity requirements. The present framework assumes a transparent payload, while its extension to regenerative D2C architectures remains future work.

Against this architectural backdrop, recent 3GPP Release 17 and 18 work items, a D2C service concept envisions satellites communicating *directly* with unmodified consumer smartphones and standard, off-the-shelf IoT devices [2]. This eliminates the intermediary ground gateway entirely, enabling true "out-of-the-box" global connectivity.

While the architectural simplification of D2C is a significant leap forward, closing a link budget directly with a 23 dBm IoT device from space sits at the edge of physics. To bridge this large spatial gap, single-tier LEO constellations rely on narrowband waveforms (e.g., 180 kHz Physical Resource Blocks for NB-IoT) combined with highly aggressive link budget margins [3]. However, this extreme power spectral density concentration throttles data rates to mere kilobits per second.

To support varying latency, capacity, and coverage requirements simultaneously, the industry is transitioning towards multi-tier D2C architectures [10].By deploying multiple LEO constellations at different altitudes, for instance, Tier-1 at 350 km to 550 km for ultra-low latency, and Tier-2 at 1200 km for broad spatial coverage. Hence, the network operators can distribute the RF burden across different orbital planes. Yet, this vertical densification introduces severe cross-tier coordination challenges.

The Physics of LEO and Channel Impairments: The transition to D2C is exponentially complicated by the orbital mechanics of LEO constellations. Unlike GEO satellites, which remain stationary relative to the Earth's surface, a LEO satellite at an altitude of 600 km travels at an orbital velocity of approximately 7.56 km/s. This extreme mobility introduces severe Doppler shifts. For a D2C link operating at 2 GHz carrier frequency, the maximum Doppler shift can exceed ± 50 kHz as the satellite rises, peaks, and sets. By contrast, standard terrestrial LTE oscillators are designed to tolerate frequency offsets of less than 1 kHz [5].

Furthermore, D2C systems suffer from highly variable timing uncertainties. A terrestrial cell tower expects a response from a UE within a tightly bound Round-Trip Time (RTT) of approximately 1 ms. In a LEO-D2C network, propagation delays stretch the RTT to between 10 ms and 40 ms, far exceeding standard terrestrial Timing Advance limits [2]. D2C networks must implement robust pre-compensation mechanisms to maintain uplink synchronization. Finally, as the satellite sweeps across the sky, the elevation angle dictates the LOS probability, forcing the signal to rapidly transition between Rician fading (direct path) and severe Rayleigh fading (obstructed by urban canyons).

The Spectrum Bottleneck and Regulatory Reality: The final, and perhaps most daunting, hurdle in the D2C evolution is spectrum availability. To communicate with unmodified terrestrial devices, D2C satellites cannot use dedicated Ku/Ka satellite bands; they must operate within existing, highly congested mobile spectrum (e.g., 1.9 GHz uplink and 2.6 GHz downlink) [6]. This introduces a heavy spatial spectrum constraint. Terrestrial operators have optimized these sub-6 GHz bands for ground networks with cell radii of 1 km to 5 km. Conversely, a single LEO satellite beam footprint covers a radius of 50 km to 500 km. When a D2C satellite transmits over a dense urban area, it floods a zone potentially containing 10^5 terrestrial IoT devices per square kilometer. Injecting satellite signals into these exact frequency bands creates a precarious co-channel interference environment, forcing D2C mega-constellations to conflict directly with hundreds of terrestrial macro-cells and competing orbital tiers.

The Need for a Spectrum Intelligence Layer: First, massive IoT deployments are physically crippled by strict 3GPP standards, which cap their maximum transmit power at a mere 23 dBm (200 mW) [11]. Second, D2C networks are forced to operate within the regulatory confines of shared, highly congested sub-6 GHz spectrum. Finally, these fragile

TABLE I: Architectural Evolution: Traditional NTN, Single-Tier D2C, and Proposed Multi-Tier D2C Ecosystems

Parameter	Traditional NTN (Relay-Centric)	Single-Tier D2C	Proposed Multi-Tier D2C w/ Spectrum Intelligence
Architecture & Hardware Requirements			
Network Architecture	"Bent-pipe" relay: Sat $\rightarrow$ Gateway $\rightarrow$ UE.	Direct access: Sat $\rightarrow$ UE/IoT. Single orbital plane.	Direct access across multiple layered orbital planes (e.g., Tier-1 at 550 km, Tier-2 at 1200 km).
Ground Hardware	Requires specialized high-gain gateways (e.g., VSAT).	Unmodified standard UE / off-the-shelf IoT devices.	Unmodified standard UE / off-the-shelf IoT devices.
Antenna & Power (Ground Device)	Sophisticated tracking antennas ($>$ 30 dBi), 10 W – 50 W.	Zero-gain omnidirectional antennas, max 23 dBm (200 mW).	Zero-gain omnidirectional antennas, max 23 dBm (200 mW).
Link Budget & Performance			
Waveforms & Data	Broadband data rates; path loss ($>$ 160 dB) easily compensated.	Narrowband (e.g., 180 kHz PRBs); throttled to kilobits/sec.	Narrowband, but overall capacity scales via cross-tier spatial diversity.
Coverage & Link Performance	Fails to support pervasive, massive IoT deployment.	Severe trade-off: Fails to support ultra-low latency and global coverage simultaneously.	Tier-1 provides lower path loss and higher SE, while Tier-2 provides wider geographic coverage.
Resource Management	Static frequency allocation via gateway scheduling.	Rigid, high-latency satellite-side scheduling.	Dynamic, ground-based *Spectrum intelligence layer* executing active connection steering.
LEO Physics & Channel Impairments			
Doppler Shift (2 GHz)	Negligible (GEO) or easily compensated by gateway hardware.	Extreme ($\pm$50 kHz); strains standard LTE oscillator tolerances.	Extreme, but intelligently predicted and managed by the ground-controller layer prior to connection.
Propagation Delay	Static ($\approx$ 250 ms for GEO).	Highly variable RTT (10 – 40 ms); exceeds standard LTE limits.	Highly variable RTT; propagation delay is modeled as a link characteristic but is not an optimization objective.
Spectrum Bottleneck & Coexistence			
Spectrum Band	Dedicated satellite frequency bands (e.g., Ku/Ka band).	Shared, congested sub-6 GHz mobile bands (1.9 GHz / 2.6 GHz).	Shared, congested sub-6 GHz mobile bands (1.9 GHz / 2.6 GHz).
Interference State	Minimal co-channel interference; localized distribution.	Precarious localized RF congestion; high inter-satellite interference.	Congestion is actively bypassed via smart spectrum sharing and dynamic tier association.

links must be continuously synchronized across multiple fast-moving LEO satellites and tiers. At the precise intersection of these extreme hardware limitations, regulatory boundaries, and orbital mechanics, traditional static spectrum management fundamentally collapses. Static frequency allocation cannot dynamically resolve the interference and capacity constraints of overlapping 500 km satellite footprints.

The network architecture must evolve. To sustain the D2C revolution, the connected devices themselves are envisioned to form a ground-controller-assisted *spectrum intelligence layer*. Rather than relying on rigid, high-latency scheduling from the satellite payload, this ground-based cognitive layer must sense local user density, predict Doppler and timing impairments, and intelligently steer connections across the available tiers in real-time. By executing optimized, device-driven spectrum sharing, the intelligence layer provides the only viable pathway to scale the spectral efficiency of D2C IoT networks.

III. Proposed Framework: Spectrum Intelligence for Multi-Tier D2C Networks

Our proposed framework replaces the existing mostly static mapping of devices to satellites with a dynamic, ground-controller-assisted orchestration model that actively optimizes resource-normalized SE across the shared D2C bandwidth.

The considered system serves two heterogeneous device classes: broadband UEs with relatively high data-rate requirements and IoT devices with lower power, intermittent activity, and narrowband traffic demands.

A. Ground-Controller-Assisted Orchestration and the Multi-Tier Model

The framework operates over a multi-tier LEO-D2C network consisting of two distinct satellite tiers. The lower-altitude tier (e.g., 550 km) functions as the *capacity tier*, since its shorter slant range reduces propagation loss and delay, resulting in a stronger link budget, higher SINR, and greater per-device spectral efficiency. The higher-altitude tier (e.g., 1200 km) acts as the *coverage tier*, as its larger satellite footprint and extended visibility region enable broader geographic coverage and improved service continuity over remote, sparsely served, or locally congested areas. A defining feature of this architecture is its ground-controller decision layer. Neither the UEs nor the IoT devices are tasked with executing independent constellation-selection algorithms. Instead, the ground controller acts as an orchestrator, absorbing all computational burdens associated with channel-condition estimation, load monitoring, and assignment optimization. This design is highly critical for massive IoT deployments, as it strictly preserves the limited power, processing capability, and signaling overhead of simplistic IoT devices.

The ground gateway provides the feeder-link interface between the satellite network and the terrestrial core and for-

wards user and control traffic. In contrast, the ground controller performs tier association, resource coordination, and handover decisions. The two entities are logically distinct but may be physically co-located.

The assumed ground-based controller placement is consistent with the transparent-payload NTN architecture considered in 3GPP TR 38.821 [2], in which the gNB functions reside on the ground and the satellite acts as a relay.

B. The Two-Stage Spectrum Intelligence Protocol

To effectively coordinate connectivity across the multi-tier ecosystem, the intelligence layer periodically executes a two-stage association protocol at every decision interval:

1) **Intelligent Constellation (Tier) Selection:** First, the ground controller evaluates a composite metric matrix for every active UE and IoT device. This matrix accounts for the estimated channel condition, current constellation load, localized IoT activity bursts, outage risk, and cross-tier handover costs. Based on these parameters, the intelligence layer assigns the device to either the capacity tier or the coverage tier.
2) **Greedy Satellite Selection:** Once a device is assigned to an orbital tier, the framework executes a localized greedy selection algorithm. The ground controller identifies and connects the device to the specific satellite within that chosen constellation that yields the maximum SINR. Crucially, this greedy optimization is applied symmetrically, maximizing SINR for both uplink telemetry and downlink command transmissions.

Association Decision Logic. The ground controller evaluates each device–tier pair using a metric that combines link quality, current tier load, outage risk, and handover cost. Devices are then sequentially assigned to the feasible tier with the higher metric, with tier loads updated after each assignment. Following tier selection, the visible satellite providing the highest SINR is selected within the assigned tier.

Complexity, Scalability, and Signaling Overhead. Based on the above sequential association procedure, the computational and signaling requirements can be characterized as follows. Let K, S, $\bar{S}_v$, N, and N_a denote the numbers of tiers, satellites, visible candidates per device and tier, total devices, and devices participating in association, respectively. Satellite propagation, link estimation, and greedy association scale as $\mathcal{O}(S)$, $\mathcal{O}(NK\bar{S}_v)$, and $\mathcal{O}(N_a \log N_a + KN_a)$, respectively. Thus, constellation size affects satellite propagation and candidate-link evaluation, while user density and the number of tiers affect link estimation and association complexity. Telemetry overhead scales as $\mathcal{O}(KN_a)$, whereas association signaling scales linearly with the number of served devices. Link calculations can be parallelized, while telemetry collection, backhaul transport, processing, and signaling delays must remain below the 10-second association interval.

C. Physics-Informed Spatiotemporal Channel Pipeline

The foundation of the two-stage protocol is a rigorous spatiotemporal channel modeling pipeline. Because LEO satellites travel at extreme velocities, the physical channel parameters are highly ephemeral.

The parameters are estimated separately for each access link between a UE or IoT device and a satellite, rather than from the satellite to ground controller link. Device location and satellite ephemeris identify visible satellites and determine link geometry. The ground controller then evaluates propagation loss, fading, interference, and SINR for each candidate link and uses these estimates for tier and satellite association. The pipeline continuously feeds the intelligence layer with real-time link estimates by capturing four critical physical effects:

- **Orbit Propagation and Geometry:** The model continually computes the Earth-Centered Earth-Fixed (ECEF) coordinates and velocities of satellites across all tiers, calculating the instantaneous slant range, elevation angle, and visibility windows for every ground device.
- **Doppler Shift and Path Loss:** The pipeline actively calculates the severe Doppler shifts induced by LEO trajectories and evaluates large-scale propagation losses. This incorporates fundamental free-space path loss alongside elevation-dependent tropospheric and ionospheric signal degradation.
- **Stochastic Fading and Atmospheric Attenuation:** To capture the stochastic nature of the D2C channel, the model integrates elevation-dependent LOS probabilities, rapid transitions between Rician and Rayleigh small-scale fading, and log-normal shadowing. A simplified ITU-R P.618/P.838-based rain attenuation model is also utilized to account for time-varying atmospheric blockages.
- **Interference and Cross-Service Leakage:** Finally, the pipeline aggregates co-channel inter-satellite interference and cross-service leakage between heavily overlapping UE and IoT transmissions. These interference metrics are combined with thermal noise floors to generate an accurate, instantaneous SINR calculation.

For each device and candidate satellite, the model derives the instantaneous slant range, elevation angle, and radial velocity from the orbital geometry. These quantities determine the free-space path loss and Doppler shift. The received power further incorporates tropospheric and ionospheric losses, rain attenuation, correlated shadowing, and LOS-dependent Rician or Rayleigh fading. The resulting SINR accounts for thermal noise, co-channel satellite interference, and cross-service leakage. The corresponding orbital, link-budget, receiver, traffic, and modulation parameters used for the performance evaluation are presented in Section IV-A.

Direction Dependent Link Budget Constraints. The framework supports both uplink and downlink communications. In the uplink, limited IoT transmit power reduces the link margin to the higher altitude tier, while narrowband transmission, robust coding, repetitions, and satellite receive gain improve link feasibility. In the downlink, the link margin depends mainly on satellite EIRP, propagation loss, and IoT receiver characteristics. The spectrum intelligence layer assigns a device to the 1200 km tier only when its estimated link quality satisfies the required threshold; otherwise, it selects the 550 km tier.

D. Dynamic Optimization and Spectrum Sharing

The ultimate objective of the spectrum intelligence layer is not merely to guarantee the absolute best instantaneous link for every single device, but to improve the resource-normalized UE-IoT spectral efficiency.

For each device, resource-normalized SE is computed by multiplying its SINR-dependent MCS spectral efficiency by the class-specific resource fraction and activity status, and then dividing the result by the number of active devices of the same class sharing the assigned tier resources. The metric therefore captures both link quality and resource congestion.

Accordingly, the spectrum intelligence layer improves this metric by treating dynamic tier allocation as a load-aware spectrum-sharing problem. For broadband UEs, the tier decision heavily weighs the number of competing UEs currently sharing the wideband resource pool, prioritizing capacity distribution and handover minimization. For IoT devices, the decision logic is highly specialized: it accounts for the intermittent nature of IoT activity, narrowband resource scaling, and low-SINR avoidance. Because inactive IoT devices do not consume resources during deep-sleep cycles, the ground dynamically reclaims those spectrum blocks. When an IoT device awakens, it is instantly routed to the tier that provides the optimal tradeoff between link stability and resource availability. By actively exploiting the complementary physical characteristics of the capacity and coverage tiers, the intelligence layer successfully mitigates localized RF bottlenecks and improves equitable, high-efficiency spectrum sharing across the entire 6G ecosystem.

The resource-normalized spectral efficiency of device d from class $x \in \{\mathrm{UE}, \mathrm{IoT}\}$ assigned to tier c is defined as

$$\eta_{d,c}^{\mathrm{RN}}(t) = \frac{a_d(t)\rho_x \eta_{\mathrm{MCS}}(\gamma_{d,c}(t))}{\max\left\{1, N_{x,c}^{a}(t)\right\}}. \tag{1}$$

where $a_d(t)$ denotes device activity, ρ_x is the class-specific resource fraction, $\gamma_{d,c}(t)$ is the instantaneous SINR, and $N_{x,c}^{a}(t)$ is the number of active same-class devices sharing tier c. Thus, the metric jointly captures link quality and resource-sharing congestion."

IV. Performance Evaluation: Validating the Spectrum Intelligence Layer

To validate the efficacy and the potential of the proposed framework, we conduct system-level orbital and link-budget simulations. The primary objective is to definitively assess whether dynamic, network-driven device-to-tier association outperforms traditional static tier-association models in maximizing resource-normalized SE under severely constrained D2C bandwidths.

The evaluation focuses on data-plane spectral efficiency and does not explicitly model telemetry collection, association signaling, or handover overhead. These implementation-dependent effects will be considered in future control-plane-aware evaluations.

A. Simulation Environment and Parameters

The simulation environment represents a constrained multi-tier LEO-D2C downlink scenario serving a heterogeneous deployment of broadband UEs and NB-IoT-type devices. As detailed in Table II, the network topology consists of a 550 km capacity tier and a 1200 km coverage tier. The geographic service area is modeled as a 20 km $\times$ 20 km urban region centered around Tokyo, Japan, at approximately 35.0°N latitude and 139.0°E longitude. To reflect realistic satellite geometry, satellite motion is generated using real TLE-based orbital propagation, enabling real satellite coordinates, slant range, elevation angle, visibility, and Doppler conditions to be evaluated for each ground device.

The simulated access network operates over a 2 GHz carrier with a limited total system bandwidth of 1.92 MHz. The deployment contains 50 broadband UEs and 100 IoT devices, where IoT devices follow an intermittent activity model.

Specifically, each IoT device follows an independent Bernoulli activity model with a 5% duty cycle. At each 0.5-second simulation step, a device is active with probability 0.05, resulting in approximately five active IoT devices on average. This model produces geometrically distributed inter-activation intervals with an average duration of approximately 10 s. Inactive IoT devices contribute zero carried SE and do not consume active IoT resources. Activity states are assumed to be independent across devices and timesteps; therefore, temporal, spatial, and inter-device traffic correlations are not modeled.

To reflect UE–IoT coexistence under constrained D2C spectrum, both service classes share the same spectrum allocation but use service-specific resource pools: UEs access a wideband pool, while IoT devices are constrained to a narrowband pool scaled by $\rho_{\mathrm{IoT}} = 0.1$. The simulation runs over a 30-minute orbital window with a 0.5-second physical-layer resolution, while the spectrum intelligence layer updates device-to-tier association every 10 seconds.

TABLE II: Key Simulation and Link Budget Parameters

Parameter	Value
Simulation Parameters	
Carrier Frequency (f_c)	2 GHz
Bandwidth (B)	1.92 MHz
Orbital data source	TLE/SGP4-based orbital propagation
Fallback constellation	Walker Delta, 72 planes $\times$ 22 sats
Inclination	53°
Tier 1 Altitude (Capacity Tier)	550 km
Tier 2 Altitude (Coverage Tier)	1200 km
Satellite Tx Power (P_t)	10 dBW
Satellite Antenna Gain (G_t)	30 dBi
Satellite System Loss	2 dB
UE Noise Figure	7 dB
IoT Noise Figure	9 dB
Number of UEs / IoT Devices	50 / 100
IoT Resource fraction (ρ_{IoT})	0.1
Modulation Schemes	QPSK, 16QAM, 64QAM
Satellite EIRP	38.0 dBW
Slant Range (550 km / 1200 km)	812 km / 1693 km

B. Evaluated Baseline Schemes

To isolate the performance gains of the spectrum intelligence layer, we benchmark our proposed framework against three established static association policies:

1) **BL1 (Single-Tier 550 km):** All traffic is statically forced onto the lower-altitude capacity tier.
2) **BL2 (Single-Tier 1200 km):** All traffic is statically forced onto the higher-altitude coverage tier.
3) **BL3 (Naive Multi-Tier):** A rudimentary service-based split where all broadband UEs default to the 550 km capacity tier, and all massive IoT devices are offloaded to the 1200 km coverage tier.
4) **Proposed (Spectrum Intelligence):** The network dynamically assigns UEs and IoT devices across both tiers utilizing the two-stage protocol (intelligent tier selection followed by greedy SINR optimization) to actively balance spectrum load.

C. Link-Level Reliability and Resource-Sharing Congestion

We first analyze the received SINR distribution to characterize the link quality of the evaluated tier-association schemes. Fig. 2a shows the SINR CDF across UE and IoT device-time samples.

BL1 and BL3 provide favorable UE SINR because both assign UEs to the 550 km tier, which has lower path loss. BL2 operates at lower SINR due to the larger path loss of the 1200 km tier. The proposed method gives an intermediate UE SINR distribution because it does not maximize instantaneous SINR alone; it also considers tier load, IoT activity, outage risk, and handover cost. For IoT devices, the proposed method does not uniformly dominate BL3 in SINR, which indicates that SINR alone does not explain the final SE gain.

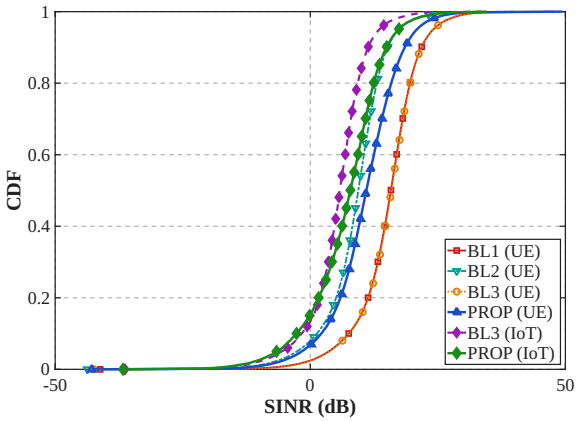
To capture the resource-sharing effect, Fig. 2b shows the normalized per-device congestion CDF. For UEs, congestion is the fraction of UEs sharing the same tier. For IoT devices, it is the fraction of active IoT devices sharing the same tier at each timestep.

The static baselines concentrate devices on fixed tiers: BL1 and BL2 place all UEs on one tier, while BL3 assigns all UEs to 550 km and all active IoT devices to 1200 km. Therefore, these schemes experience maximum normalized congestion. In contrast, the proposed method distributes UEs and active IoT devices across both tiers, shifting the congestion CDF leftward. This reduction in same-service congestion increases the effective per-device resource share and explains why the proposed method improves SE even when the SINR gain is moderate.

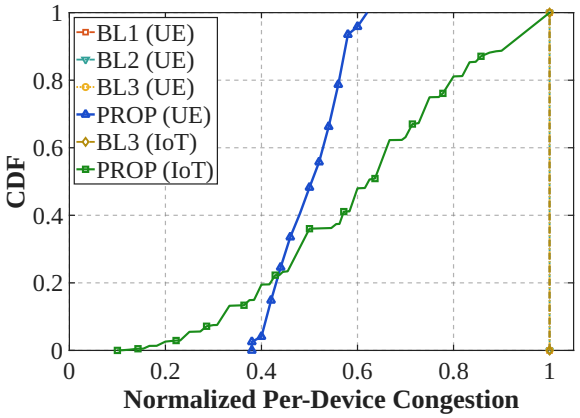
D. Network-Level Capacity: Spectral Efficiency Gain

While SINR dictates raw link quality, the ultimate metric for multi-tier D2C viability is resource-normalized SE. If a tier is heavily overloaded, a high SINR is useless because the device cannot access enough bandwidth.

Fig. 3 illustrates the profound impact of the intelligence layer on per-device average SE. For broadband UE traffic, the static BL1 and BL3 schemes cap at approximately



(a) SINR CDF for UE and IoT devices under the evaluated tier-association schemes.



(b) Normalized per-device congestion CDF, where UE congestion is defined as the fraction of UEs sharing the same tier and IoT congestion is defined as the fraction of active IoT devices sharing the same tier at each timestep. A leftward shift indicates lower resource-sharing congestion and a larger effective resource share per device.

Fig. 2: Joint CDF comparison of link quality and resource-sharing congestion.

0.0421 bps/Hz/user and 0.0419 bps/Hz/user, respectively, while the path-loss-heavy BL2 drops to 0.0275 bps/Hz/user. In stark contrast, the proposed dynamic framework achieves an average UE SE of **0.0641 bps/Hz/user**—an approximate 53% capacity increase over the best static baseline.

A similar proportional leap is observed in the massive IoT ecosystem. While absolute IoT SE is inherently lower due to severe narrowband constraints and deep-sleep duty cycles, the proposed method elevates average IoT SE from 0.0011 bps/Hz/user (under naive BL3) to **0.0018 bps/Hz/user** (a 63% improvement). To assess statistical variability, the simulations were repeated using five random seeds. Across these realizations, the proposed framework achieves average UE and IoT SE improvements of $53.12\% \pm 1.39\%$ and $45.10\% \pm 5.92\%$, respectively, over BL3 at the 95% confidence level. The representative realization shown in Fig. 3 yields an IoT improvement of approximately 63.6%. These results show that the proposed framework consistently outperforms BL3, although the magnitude of the IoT improvement varies across spatial, traffic, and channel realizations.

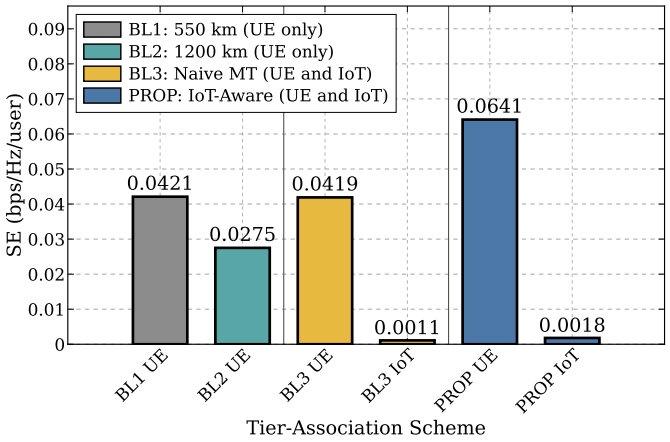


Fig. 3: Comparison of per-device average spectral efficiency. The proposed method significantly outperforms static baselines by intelligently balancing link quality against cross-tier resource loads.

Link reliability is evaluated using outage thresholds of -3 dB for visible UE links and -6 dB for active, visible IoT links. The proposed framework yields UE and IoT outage probabilities of 3.896% and 5.532%, respectively, compared with 1.286% and 4.597% under BL3. These results reflect a capacity–reliability tradeoff.

The proposed framework achieves 5th-percentile UE and IoT SE values of 0.0207 and 0.0075 bps/Hz/user, compared with 0.0175 and 0.0063 bps/Hz/user under BL3, corresponding to improvements of 18.3% and 19.0%, respectively. The IoT percentile is computed over active and visible links, whereas its average SE includes inactive periods. Packet-delay and latency distributions are not modeled and remain for future evaluation.

The proposed framework yields UE and IoT tier-switching rates of 29.97% and 0.85% per association interval, respectively, compared with 13.32% and 0.73% under BL3. The higher UE switching rate reflects the framework's dynamic response to changes in link quality and tier load. However, handover execution is treated ideally; signaling overhead, interruption time, transient SINR degradation, and packet loss during tier or satellite transitions are not explicitly modeled.

These numerical results validate our core hypothesis: intelligent, dynamic constellation tier-allocation is a strict prerequisite for 6G D2C networks. By orchestrating spectrum sharing at the network level based on load, outage risk, and channel state, the proposed spectrum intelligence layer exponentially increases spectral efficiency without requiring hardware modifications to ground devices.

Scalability Analysis. To evaluate denser deployments, additional simulations were conducted with up to 300 UEs and 1000 IoT devices while retaining the same TLE-based orbital configuration. The proposed method consistently outperforms BL3 across all evaluated scales, with UE/IoT SE gains of (42.1%, 33.3%), (33.7%, 50.0%), and (20.6%, 13.7%) for $(N_{\mathrm{UE}}, N_{\mathrm{IoT}}) = (100, 300)$, $(200, 500)$, and $(300, 1000)$, respectively. Although per-device SE and relative gains decrease at high densities due to increased resource sharing under the fixed 1.92 MHz bandwidth, the proposed framework maintains a positive gain over BL3. The TLE-based propagation also preserves the time-varying set of visible candidate satellites considered during association.

V. Future Directions and Open Challenges

While the proposed spectrum intelligence layer demonstrates significant gains in resource-normalized spectral efficiency, fully realizing this collaborative, device-driven vision on a global scale introduces formidable engineering challenges. The evolution from localized multi-tier demonstrations to pervasive 6G connectivity requires concerted research efforts across several critical vectors.

- **Collaborative Edge AI and Multi-Agent Optimization:** Executing complex spatiotemporal machine learning algorithms traditionally requires centralized computational power. However, to decentralize the intelligence layer, ground devices must collaboratively process channel modeling data and execute multi-tier routing decisions at the edge [12]. Future research should pivot toward ultra-lightweight distributed paradigms such as Multi-Agent Reinforcement Learning (MARL) and Graph Neural Networks (GNNs). Since satellite topology is well-known, a-priori associations of geolocations to tiers and satellites can also be pre-computed, allowing UEs and IoT devices to perform simple lookups from pre-calculated association tables. This can then be augmented with multi-agent ML methods to complete optimized associations.
- **Inter-Constellation Coexistence and OISL Integration:** The orbital spectrum landscape is becoming highly fragmented as competing mega-constellations deploy overlapping services. As these constellations scale, they inevitably cross paths over dense geographic regions, exacerbating co-channel interference in sub-6 GHz bands. Preventing catastrophic RF collisions requires intelligence-driven dynamic spectrum access protocols capable of negotiating *inter-network* coexistence. Furthermore, future architectures must integrate Optical Inter-Satellite Links (OISL) into the intelligence layer's decision matrix [13]. Rather than forcing a failing direct RF link through crippling interference, the intelligence engine should dynamically offload terrestrial traffic via OISL to a satellite with an unobstructed link. Future work should also examine protocol-level integration with 3GPP NTN control procedures and coexistence among independently operated constellations through inter-operator coordination, spectrum-sharing policies, and interference-management mechanisms.
- **ISAC and Massive MIMO for Rapid Beam Alignment:** To fully capitalize on the spectrum intelligence layer, future D2C satellites must integrate Massive MIMO arrays. However, unlike terrestrial base stations, LEO satellites travel at orbital velocities, making it incredibly difficult to maintain precise, narrow beams on specific clusters of IoT devices. Future solutions should explore the convergence of Integrated Sensing and Communication (ISAC) within multi-tier NTN environments [14]. By utilizing multi-modal sensing data to preemptively track device clusters and counteract rapid orbital orientation changes, ISAC-enabled beamforming can ensure that the spatial load-balancing capabilities dictated by

the spectrum intelligence layer are executed with pinpoint accuracy.

• **Predictive Cross-Tier Handover Management:** Due to the extreme mobility of LEO constellations, the visibility window of a single satellite over a geographic point is remarkably short. In dense multi-tier deployments, millions of devices may require continuous synchronized handovers, not only laterally between satellites in the same orbit but also vertically across different tiers. When the intelligence layer dynamically alters the spectrum-sharing landscape to maximize efficiency, the handover topology becomes exponentially more complex. Open research directions include "make-before-break" predictive handover protocols that anticipate satellite trajectories and preemptively migrate IoT traffic across tiers without inducing signaling storms or dropping critical telemetry. Future work should also evaluate the load-aware framework under traffic hotspots and satellite failures, where the ground controller can redirect devices from congested tiers or unavailable satellites to alternative visible links. While the current framework optimizes resource-normalized spectral efficiency, future work will evaluate control-plane RTT, handover interruption, and packet-level delay to characterize the latency impact of dynamic cross-tier association.

• **Integration of High-Fidelity 3D Orbital Digital Twins:** Finally, the static channel modeling pipelines utilized in current D2C frameworks must evolve into fully interactive, high-fidelity Digital Twins [15]. By establishing a continuous, bi-directional data exchange between the physical multi-tier constellations and a virtual replica hosted at the network edge, operators can empower the spectrum intelligence layer with vastly superior predictive models. Future research should focus on incorporating real-time ray-tracing and localized atmospheric mapping into the digital twin. This will allow the intelligence layer to simulate the exact RF impact of an unexpected surge in urban device density *before* the satellites pass overhead, validating aggressive multi-tier routing strategies in the virtual domain to ensure zero-fault execution in the physical world.

• **Optimality Limitation:** The tier-assignment problem is related to load-aware network utility maximization and generalized assignment. Since each device utility depends on tier load, link quality, outage risk, and handover cost, the decisions are coupled and exact optimization is combinatorial. The proposed sequential greedy method provides a low-complexity feasible solution without optimality guarantees. Future studies should perform comparisons with optimal solutions for small-scale scenarios

VI. Conclusion

The insatiable demand for all-pervasive connectivity has pushed traditional, single-tier spectrum allocation to its breaking point. To overcome these constraints, this article proposed a paradigm shift towards a multi-tier D2C satellite network driven by a ground-based intelligence layer. By transforming the connected IoT devices into an active, collaborative cognitive mesh, the network can intelligently route connections across different orbital tiers based on real-time channel conditions and user density. When UEs and IoT devices connect to the same constellation, this intelligence layer orchestrates optimized spectrum sharing. Through rigorous system-level simulations, we demonstrated that utilizing a spectrum intelligence layer for multi-tier D2C networks successfully bypasses localized RF congestion, normalizes throughput in highly dense environments, and exponentially increases the overall spectral efficiency. Ultimately, empowering the IoT devices to act as the intelligence layer will transform orbital constraints into unprecedented opportunities for global, resilient 6G connectivity.

Acknowledgment

This work was supported in part by U.S. National Science Foundation award 2115215.

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